SAFWAT ALI KHAN

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RESEARCH INTEREST

Software Testing, Verification and Validation, Mobile UI Testing, AI Agentic Models, Large Language Models for Software Engineering, Software Accessibility, Automated Program Repair, Trustworthy AI.

EDUCATION

George Mason University *September 2019 - Present*
PhD in Computer Science CGPA: 3.88/4.00
MS in Computer Science *(2025)* CGPA: 3.85/4.00
University of Dhaka *January 2010 - May 2014*
BSc in Computer Science & Engineering

GRADUATE RESEARCH EXPERIENCE

Current projects:

- A large-scale study on accessibility program repair and a benchmark for AI agents (manuscript currently under review).
- Conducting a systematic literature review of developer-facing tools for software accessibility, surveying development guidelines, software-enhancement techniques, targeted impairment populations, input types, experimental methodologies, evaluation metrics, and the SE lifecycle stages and tasks impacted.

Publications:

- "Testing Practices, Challenges, and Developer Perspectives in Open-Source IoT Platforms" - Daniel Rodriguez-Cardenas, **Safwat Ali Khan**, Prianka Mandal, Adwait Nadkarni, Kevin Moran, Denys Poshyvanyk in ICST 2025.
- "Advancing Mobile UI Testing by Learning Screen Usage Semantics" - **Safwat Ali Khan** in ICST 2025.
- "AURORA: Navigating UI Tarpits via Automated Neural Screen Understanding" - **Safwat Ali Khan**, Wenyu Wang, Yiran Ren, Bin Zhu, Jiangfan Shi, Wing Lam, Kevin Moran in ICST 2024.
- "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization" - Junayed Mahmud, Nadeeshan De Silva, **Safwat Ali Khan**, Seyed Hooman Mostafavi, S M Hasan Mansur, Oscar Chaparro, Andrian Marcus, Kevin Moran in ICSE 2023.
- "AVGUST: Automating Usage-based Test Generation from Videos of App Executions" - Yixue Zhao, Saghar Talebipour, Kesina Baral, Hyojae Park, Leon Yee, **Safwat Ali Khan**, Yuriy Brun, Nenad Medvidovic, Kevin Moran in ESEC/FSE 2022.

GRADUATE TEACHING EXPERIENCE

Teaching Assistant for:

- Mobile Application Development
- Computer Programming for Engineers
- Intro. to Low-level Programming
- Database Systems
- Data Structures

Responsibilities: Mentored undergraduate and graduate students, graded assignments, and led laboratory sessions. As the teaching assistant for Mobile Application Development, supervised students throughout hands-on lab hours and provided individualized guidance in diagnosing and resolving a broad range of development challenges, from debugging and UI implementation to application design.

RESEARCH TALKS

"Leveraging Machine Learning to bypass Android UI tarpits" - Fraunhofer, USA (2023)
"Automated Exploration of Android Apps" - SE Seminar, George Mason University (2022)

ACADEMIC SERVICE

Journal Reviewer

· IEEE TSE 2026

Co-Reviewer

· FSE 2024, ICSE 2023, SANER 2023, ASE 2022

Volunteer

· ICST 2024

FELLOWSHIP

Provost Scholarship - 2026
Sustainability Summer Research Fellowship - 2024
CS Research Initiation Award - 2020

WORK EXPERIENCE

Software Engineer, QA August 2014 - July 2019

Software Quality Assurance Engineer with experience across manual and automated testing of web and mobile applications. Contributed to impactful projects including systems for underserved populations in Bangladesh, web-based learning management systems for healthcare professionals, and health tracking modules for developmentally disabled individuals.

SKILLS

Programming Languages:	Python, Java, C, C++, JavaScript
Machine Learning APIs:	CLiP, sklearn, tensorflow, pyTorch
Test Automation Tools:	Selenium Webdriver, JMeter, UIAutomator, Google Monkey, APE

MENTORSHIP

· Co-mentor for the Aspired Scientist Summer Internship Program, 2022 and 2020.

EXTRA CURRICULAR ACTIVITIES

· Treasurer for the Bangladeshi Graduate Student Association at George Mason University.

REFERENCES

Dr. Kevin Moran
· Assistant Professor of Computer Science
· University of Central Florida
· Email: kpmoran@ucf.edu

Dr. Andrian Marcus
· Professor of Computer Science
· George Mason University
· Email: amarcus7@gmu.edu

Dr. Wing Lam
· Assistant Professor of Computer Science
· George Mason University
· Email: winglam@gmu.edu

Dr. Adwait Nadkarni
· Associate Professor of Computer Science
· William & Mary
· Email: apnadkarni@wm.edu